

Course Code: CCE-332,
Course Title: Peripheral and Interfacing sessional

Group: 01

Comparison with related works:

Name	Primary Function(s)	Locomotion & Control	Key Features & Innovations
Farm Droid (our Project)	Real-time monitoring, robotic manipulation, and weeding	Rover, Raspberry Pi 4 (brain), Arduino (controller)	Real time monitoring with weeding, modular robot arm, as well as opencv for image detection. Alongside Gemini is used for LLM integration
Modular 4WD Robot [1]	Grass cutting, leaf crushing, and precision seeding	4WD skid-steer, LiDAR/Camera, EKF state estimation	Features a vertically stacked two-unit architecture with quick-release toolheads. Uses simulated coverage-path algorithms (like zigzag/boustrophedon) for 100% gap-free field coverage.
AgriCruiser [2]	Over-the-row precision spraying for weed management	2-wheel front differential drive with rear casters	Open-source, highly reconfigurable chassis made from T-slot extrusions. Offers adjustable track width (1.42–1.57 m) and high clearance at a low deployment cost of ~\$5,000–\$6,000.
Six-Wheeled Laser Weeder [3]	High-precision laser weed elimination	6-wheeled, double four-bar linkage suspension	Uses a 3-axis linear actuation mechanism to position a 10W blue diode laser directly over weeds, achieving a 97% hit rate while navigating obstacles up to 15 cm.
FarmDroid FD20[4]	Autonomous seeding and preventative mechanical weeding	4-wheeled, solar-powered, GPS-guided	Operates on a memory-based approach, recording exact GPS coordinates of planted seeds to mechanically weed between and within rows without complex real-time vision. Heavily studied for climate-smart soil management.
BoniRob [5][6]	Crop monitoring, targeted fertilization, and weeding	4-wheel independently steered	Adjusts working width dynamically (0.75m–2.0m) and uses mechanical cylinders to crush weeds as an alternative to herbicide spraying.
ROBOTTI 150 D [7]	Multipurpose field operations (evaluated for soil health impact)	Lightweight wheeled vehicle	Used extensively in research to quantify soil compaction. Studies show it causes less subsoil compaction than heavy tractors, though repeated wheeling still alters topsoil porosity.
Tertill [8]	Home garden weed control	Solar-powered mini rover	A consumer-grade, waterproof robot that autonomously navigates small plots and uses a simple wire/string trimmer to cut newly sprouted weeds.

References:

- [1] Kumar, A., Kamalaksha, S.A., Srividya, R., Zuber, M., Ahmad, K.A., Singh, S. and Nair, V.G., 2026. Modular 4WD agricultural robot for cutting, collection, and precision seeding: design and simulation-based evaluation. *Scientific Reports*.
- [2] Truong, K., 2025. *AgriCruiser: An Open Source Agriculture Robot for Over-the-row Navigation*. University of California, Los Angeles.
- [3] Usama, M., Khan, M.I., Hasan, A., Nadeem, M.S., Iqbal, K.F., Aslam, J., Ali, M.A. and Awan, A.N., 2025. Design of a six wheel suspension and a three-axis linear actuation mechanism for a laser weeding robot. *arXiv preprint arXiv:2512.10319*.
- [4] Bručienė, I., S. Buragienė, and E. Šarauskis. 2025. “Comparative Evaluation of the Effect of Conventional and Robotic Mechanical Weeding on Topsoil Physical Attributes.” *Soil and Tillage Research* 254: 106739. <https://doi.org/10.1016/j.still.2025.106739>
- [5] Schaefer, A., *Plant Robot “Bonirob”*, Available at: <https://www.youtube.com/watch?v=utiNto4BeOg>, (last accessed 17.06.2026).
- [6] Langsenkamp, F., F. Sellmann, M. Kohlbrecher, et al. 2014. “Tube Stamp for Mechanical Intra-Row Individual Plant Weed Cover.” *Agricultural Engineering International: CIGR Journal*: 16–19.
- [7] Calleja-Huerta, A., M. Lamandé, O. Green, and L. J. Munkholm. 2023a. “Impacts of Load and Repeated Wheeling From a Lightweight Autonomous Field Robot on the Physical Properties of a Loamy Sand Soil.” *Soil and Tillage Research* 233: 105791. <https://doi.org/10.1016/j.still.2023.105791>
- [8] Sanchez, J., and E. R. Gallandt. 2021. “Functionality and Efficacy of Franklin Robotics' Tertill Robotic Weeder.” *Weed Technology* 35: 166– 170. <https://doi.org/10.1017/wet.2020.94>